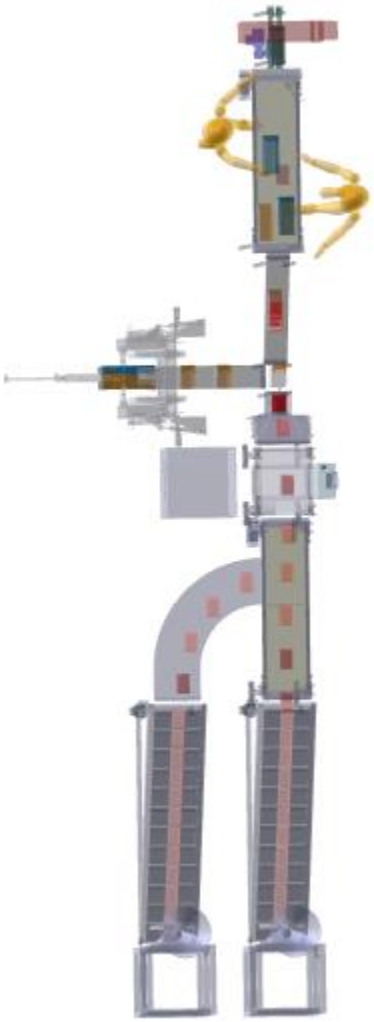
NIHVA
NIHVA TECHNOLOGIES PVT. LTD.

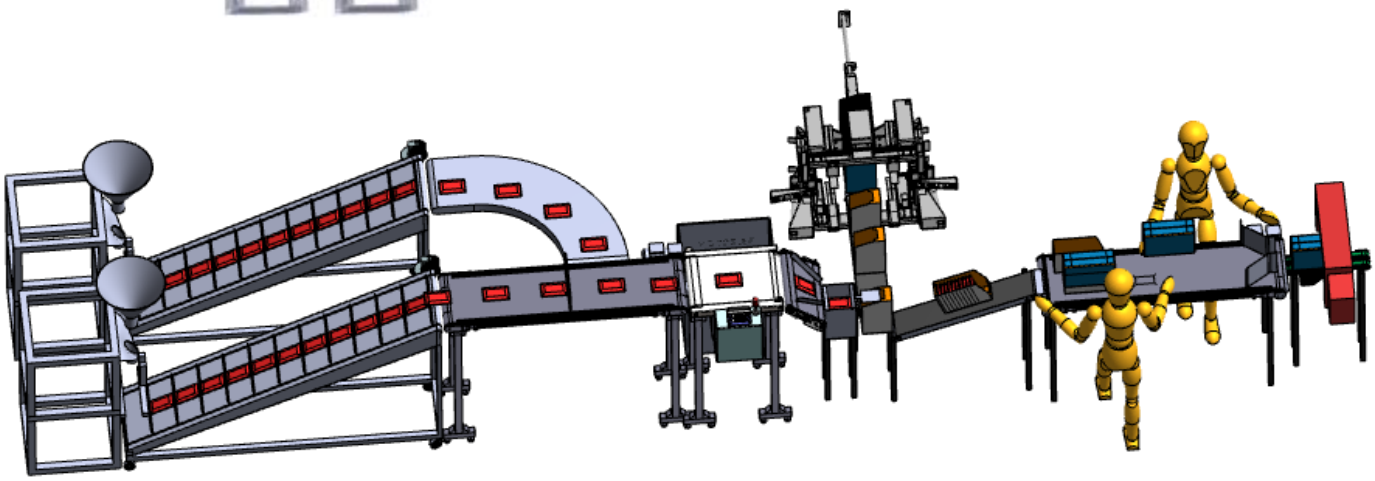
AUTOMATION SOLUTION-DIVISION



ITC Spices Automation

Proposal:

- Packing
- Sorting
- Filling
- Inserting



We Make Possible

Date: 2nd May 2017

ITC Limited
Guntur, Andhra Pradesh,

Kind Attn: Mr. Kiran Kumar

Dear Sir,

Thank you for giving us an opportunity to participate with our offer on the above project. We enclose herewith our proposal which includes:

A. Requirement
B. Scope of Works
C. Pricing
D. Technical Specification
E. Terms & conditions

A. Requirement

1. **Energy efficient Design:** The secondary packing line should be energy efficient. The design shall ensure minimum possible power.
2. **Equipment Reliability:** Downtime of the secondary packing automation should not be more than 2%. The line should ensure no choking or tare of the pouches, trays and boxes.
3. **Check weighs:** Primary packed pouches must be checked for their weights to ensure that they are within the specified limits, if they are not within the specified limits they must be diverted to slitting section.
4. **Stacking of pouches:** The accepted pouches must be stacked into specified number, which may vary depending on SKU's before being boxed. The pouches must be placed vertically into the tray. The details of SKU's and quantities have been given in annexure.
5. **Tray and lid folding:** The trays must be folded automatically and be positioned for the placement of stacked pouches. Lid has to be placed over the tray for closer.
6. **Sealing and labelling:** The trays closed with lids must be sealed with tape automatically, simultaneously the labelling must be done. The labelling should be highly accurate with minimal levels of tolerance.
7. **Feedback systems:** There must be feedback systems for the following
    - **Pouch counting:** The number of pouches accepted must be counted before being stacked, which shall be giving feedback to primary packing machine and laminate printer for stopping when required number is reached.
    - **Packing extra:** The auger count has to be controlled form the feedback provided by the check weighs, which shall be ensuring the acceptance and rejection of the pouches depending on the specified limits so that packing extra is reduced.

1. **Data collection:** Data should be gathered from all possible points of the system. Few of the main points would be like start and stop data of machinery, number of pouches being packed, number of pouches being rejected, weight variations of pouches, number of boxes being sealed. It is preferred if we can print few data online at the end of each run like start, end time and number of pouches packed.
2. **Safety:** Secondary packing automation should comply with industrial standards of safety, occupational health and hygiene as per industry norms. For example:
    - Provision of limit switches, where necessary
    - Guards for belts/ pulleys
    - Isolator switches for machine maintenance
3. **Flexibility:** The automation has to be flexible in order to accommodate variations in pouches, lids and trays in future if required without much changes from the actual system.

B.SCOPE OFWORKS:

The scope of work shall be exclusively on secondary packing. This includes collection of pouches from the VFFS packing machines, checking the weight of the pouches, stacking the pouches into tray, closing with the lid(MANUAL), sealing & labelling the box.

For the above scope, following set of activities are expected to be performed:

1. Design and development of secondary packing automation.
2. Development of site drawings for secondary packing automation.
3. Budgetary proposal for procurement, installation and commissioning of secondary packing automation with break up.
4. Project implementation plan.

I. Drawings

We will issue a full set of shop drawings for approval within 2 weeks of receiving the confirmed order and finalized design details.

Training

After installation of the automation system, we will train the Clients personnel in the operation and basic maintenance, including the understanding of the error codes on the control system (if applicable).

Warranty

The warranty period is twelve months from the date of installation and commissioning.

If a defect arises within the warranty period stated, which is attributable to faulty design, materials or installation workmanship the product will be repaired at our discretion without charge. This guarantee is subject to proper use of the product and assumes that the product is regularly serviced and maintained by skilled personnel in accordance with the operation and maintenance instructions supplied.

Maintenance and Service Agreement

We would suggest that you take advantage of a maintenance and service agreement on the Nihva products, which will guarantee trouble free operation and speed of response in an emergency. Customized proposals are available on request.

C.PRICING

A.Packaging & sorting system:

Sr No	Description	FFS	Price in INR/Each	Total Amount In INR
1	Packaging & Sorting System	No's	31,25,000	31,25,000
2	Installation & Commissioning charges for complete system	No's	2,60,000	2,60,000

B.Labelling system:

Sr no	Description	FFS	Price in INR/Each	Total Amount In INR
1	Labelling and Roll Selector	No's	6,70,000	6,70,000

Date: 2nd May 2017

D: Technical Specifications:

1	**Carton erector with 1 magazine**
	a> Small Conveyor b> Pneumatic vacuum based Pickup assembly c> Same pick up assembly for two magazines d> Capacity: - 15-20 Cartons/min e> Power Supply: - 3 ph. 440/220v 50Hz f> HMI control input g> PLC Control Panel h> Working Height: - maximum 2000mm tentative
	Rates: 6,40,000
2	**Pouch stacker conveyor with servo control**
	a) this palletizer work after check weighed b) Working speed adjustable as per check weighed speed c) Due to servo control accuracy is very high d) Food Grade belt e) SS structure
	Rates: 5,30,000
3	**Take up conveyor with VFD for 2 FFS**
	a) Take up conveyor for speed matching of check weighed and FFS b) Motor is as per Annexure c) VFD is provided to adjust speed d) Divertor selection option
	Rates: 4,60,000
4	**Carton sealer**
	a) Tape Size: - 3" b) belt Speed: - 10m /min c) Power Supply: - 3 ph. 440/220v 50Hz d) Motor is as per Annexure e) both side with pop station
	Rates: 3,90,000
5	**Pouch weighed with 120 PPM**
	a) 120 PPM adjustable b) Highly accurate in motion weigh and reject equipment is offered with several weight indicator and line speed option. c) Provides bulk total and feed rate to an accuracy of +/- 0.6 g d) capacity: - 100gms to 3000kg e) Belt speed: - 35 mtr/min as per selection of weight
	Rates: 6,10,000

6	**stacker Assembly with collecting Arm**
	a) Servo based fix rotation ARM b) Pneumatic action figure c) Action start after input signal from previous system d) figure will be change according to product size
	Rates: 4,95,000

E: Terms & Conditions:

1	**Price**	**:**	Ex Works Guntur
2	**P & F Charges**	**:**	include
3	**Freight Notes**	**:**	Include
4	**Delivery**	**:**	10-12 weeks from the date of receipt of confirmed purchase order & Advance Payment.
5	**Payment**	**:**	30 % Advance against Bank Guarantee 50 % After trial & Dispatch. 10 % against Installation & commissioning. 10 % against PBG
6	**Insurance**	**:**	Include
7	**Sales Tax**	**:**	Extra as applicable at the time of dispatch. Presently @13.5% VAT (supply within Maharashtra OR @2% CST against Form C (Outside Maharashtra) Or GST will be applicable if required.
8	**Service Tax**	**:**	Extra @ 15%
9	**Validity**	**:**	Validity of Offer is 15 days from the date of quotation
10	**Warranty**	**:**	12 months from the date of supply.

Please feel free to contact **Mr. Mayur Derle** on 7709209333 For any further Clarification /Information in this concern.

With Best Regards

Pooja Badre
7709586333
NIHVA Technologies Pvt Ltd